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探索性烹饪方法报告 • Exploratory Culinary Methods Report

小苏打浓度与煮制时间对意大利面拉面化效果的影响：一项 3 × 3 探索性厨房实验

Effects of Baking Soda Concentration and Cooking Time on the Ramen-Style Transformation of Spaghetti: A 3 × 3 Exploratory Kitchen Trial

作者 / Author: 留子也要好好吃面 • Liuzi Ye Yao Haohao Chi Mian

文章编号 / Article ID: NN-2026-001　发表 / Published: October 2026



图 1 / Figure 1.
成品示例：小苏打处理后的意大利面用于汤面或拌面式呈现。图中商品包装仅用于说明实验材料；本研究与制造商无赞助或商业关系。 / Finished serving example using baking-soda-treated spaghetti. Product packaging identifies the material only; no sponsorship or commercial relationship is claimed.

摘要与关键词 / Abstract and Keywords

摘要： 在难以获得新鲜碱水面或拉面的环境中，商品干制意大利面常被用作替代材料。本探索性厨房实验采用 3 × 3 设计，比较每升水中加入 2、4、6 g 小苏打，并分别煮制 10、12、15 分钟的九种组合。作者依据弹性、顺滑度、碱香纯度和接近拉面的程度进行结构化感官评价，并给出 10 分制综合评分。4 g/L、15 分钟处理得分最高（9.5/10），被描述为颜色金黄、Q 弹顺滑、碱香适中且无明显苦味；4 g/L、12 分钟得分 8.5/10，更适合偏硬口感。2 g/L 处理整体变化有限；6 g/L 随时间延长出现苦味、涩感、黏口和肥皂味。结果表明，在本次单人、探索性厨房条件下，4 g/L 小苏打与 15 分钟煮制是最优组合。

关键词： 碱水面；小苏打；意大利面；拉面替代；面条质构；厨房实验

Abstract: Commercial dried spaghetti can serve as a substitute when fresh alkaline noodles are difficult to obtain. This exploratory kitchen trial used a 3 × 3 design combining baking soda concentrations of 2, 4, and 6 g/L with cooking times of 10, 12, and 15 minutes. Treatments were assessed for elasticity, smoothness, alkaline aroma, and perceived similarity to ramen-style noodles, followed by an overall score on a 10-point scale. The highest score, 9.5/10, was assigned to 4 g/L for 15 minutes, described as golden, springy, smooth, appropriately alkaline, and not noticeably bitter. The 4 g/L, 12-minute treatment scored 8.5/10 and retained a firmer center. Treatments at 2 g/L produced limited transformation, whereas prolonged cooking at 6 g/L produced bitterness, astringency, stickiness, and a soap-like odor. Within the limits of a single-observer exploratory trial, 4 g/L for 15 minutes gave the most favorable result.

Keywords: alkaline noodles; baking soda; spaghetti; ramen substitute; noodle texture; culinary methods

1. 背景 / Background

海外生活或食材获取受限时，方便获得的干制意大利面可作为亚洲面条的替代基础。小苏打（碳酸氢钠）使煮面水呈碱性，并可能改变面条颜色、气味和口感。本实验旨在以结构化厨房测试比较小苏打浓度与煮制时间的组合，寻找兼顾弹性、顺滑度与适度碱香的实用条件。

When fresh Asian alkaline noodles are inaccessible, widely available dried spaghetti offers a practical starting material. Adding baking soda makes the cooking water alkaline and may alter color, aroma, and texture. The objective was to compare concentration-time combinations and identify a practical condition balancing springiness, smoothness, and moderate alkaline character.

2. 材料与方法 / Materials and Methods

材料 / Materials. 商品干制硬质小麦意大利面、小苏打和饮用水。每组使用 1 L 水；原帖未记录每组干面质量。商品由作者自行购买。 / Commercial dried durum-wheat spaghetti, baking soda, and water were used. Each treatment used 1 L water; the dry spaghetti mass per treatment was not documented. The product was independently purchased.



图 2 / Figure 2. 实验材料与 4 g 小苏打称量。 / Commercial dried spaghetti and measurement of 4 g baking soda.

实验设计 / Design. 小苏打浓度为 2、4、6 g/L；煮制时间为 10、12、15 分钟，共九种组合。水量统一为 1 L，中火煮制。原帖提及普通意大利面作为概念性对照，但未提供独立对照评分。 / Baking soda concentrations were 2, 4, and 6 g/L, combined with cooking times of 10, 12, and 15 minutes. Each treatment used 1 L water over medium heat. Ordinary spaghetti was mentioned as a conceptual control, but no separate control score was supplied.

评价 / Evaluation. 作者从弹性、顺滑度、碱味纯度和接近拉面的程度四方面进行主观评价，并给出 10 分制综合评分。未进行盲评、重复试验或仪器测量。 / The author evaluated elasticity, smoothness, alkaline-aroma quality, and ramen-like similarity, then assigned an overall 10-point score. No blinding, replication, or instrumental testing was reported.

3. 结果 / Results

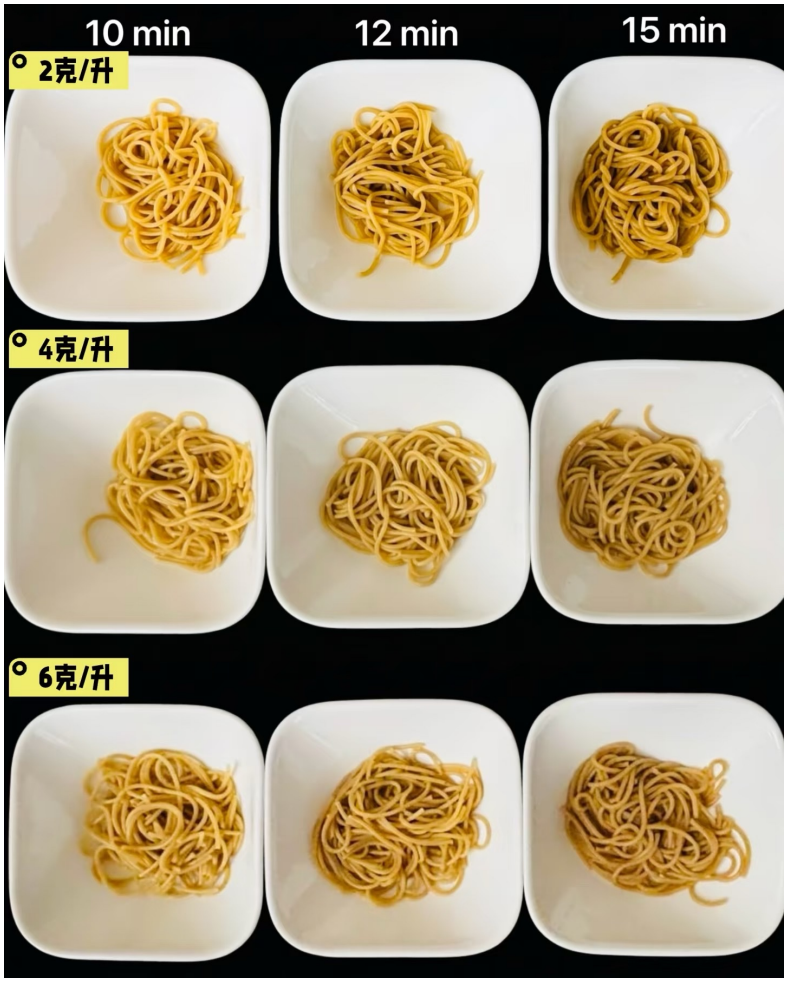


图 3 / Figure 3. 九种小苏打浓度与煮制时间组合的外观对比。图像显示碱浓度和煮制时间增加时颜色总体加深，但照片未经过颜色仪校准。 / Visual comparison of nine treatments. Color generally darkened with concentration and time, but images were not instrumentally color-calibrated.

表 1 / Table 1. 九种处理的综合感官评分（10 分制） / Overall descriptive scores for the nine treatments (10-point scale).

小苏打 Baking soda	10 min	12 min	15 min
2 g/L	5.0	6.0	6.5
4 g/L	7.0	8.5	9.5
6 g/L	6.0	5.0	3.5

详细观察 / Detailed Observations

处理 / Treatment	主要观察 / Main observation
2 g/L • 10 min	5.0：仍像意面，有硬芯，几乎无碱香。 / Still spaghetti-like; firm core; almost no alkaline aroma.
2 g/L • 12 min	6.0：弹性一般，碱香很淡，拉面感弱。 / Average elasticity; faint alkaline aroma; weak ramen character.
2 g/L • 15 min	6.5：熟透但偏软，筋道不足。 / Fully cooked but soft and insufficiently springy.
4 g/L • 10 min	7.0：外弹内硬，中心仍有白芯。 / Elastic exterior but firm, pale center.
4 g/L • 12 min	8.5：接近拉面，很弹，芯略硬。 / Close to ramen; very elastic; slightly firm center.
4 g/L • 15 min	9.5：最像拉面，Q 弹顺滑，碱香适中。 / Most ramen-like; springy, smooth, balanced alkaline aroma.
6 g/L • 10 min	6.0：碱味偏重、芯硬、回口微苦。 / Strong alkalinity; firm center; mildly bitter finish.
6 g/L • 12 min	5.0：苦涩，仍有弹性但黏口。 / Bitter and astringent; elastic but sticky.
6 g/L • 15 min	3.5：肥皂味、表面发黏、苦味明显。 / Soap-like odor, sticky surface, pronounced bitterness.



图 4 / Figure 4. 原始评分矩阵和中文描述。 / Original score matrix and Chinese treatment notes.

4. 实用方法 / Practical Procedure

步骤 / Step	操作 / Procedure
1	向 1 L 水中加入 4 g 小苏打并加热至沸腾。 / Add 4 g baking soda to 1 L water and bring to a boil.
2	水开后加入意大利面。 / Add spaghetti after the water boils.
3	煮满 15 分钟；偏好更硬口感者可尝试 12 分钟。 / Cook for 15 minutes; 12 minutes may suit a firmer preference.
4	使用容量较大的锅并留意泡沫。 / Use a larger pot and monitor foaming.
5	捞出后用于汤面或拌面；根据菜式调整调味。 / Drain and use in soup or mixed-noodle dishes; season to suit the preparation.



图 5 / Figure 5. 碱性煮面水出现泡沫和黄褐色。该图仅记录外观；实验未测量 pH、色差或煮制损失。 / Alkaline cooking water developed foam and a yellow-brown color. Appearance only; pH, color difference, and cooking loss were not measured.

5. 讨论 / Discussion

评分呈现明显的浓度与时间组合效应：2 g/L 随时间增加仅小幅改善；4 g/L 随时间增加持续改善；6 g/L 则随时间增加而下降。最优点出现在中等浓度与最长煮制时间，而高浓度长时间处理产生明显不良气味和口感。该模式说明“更多碱”并不等同于“更像拉面”，碱浓度与面芯熟化需要共同平衡。

The score pattern suggests a concentration-time interaction. Scores rose modestly over time at 2 g/L, rose strongly at 4 g/L, and declined over time at 6 g/L. The best outcome occurred at the middle concentration and longest cooking time, whereas prolonged high-alkali treatment produced undesirable flavor and surface

properties. More alkali did not necessarily create a better ramen-like result; concentration had to be balanced with center softening.

6. 局限性 / Limitations

本研究属于单人探索性厨房实验。原始材料未报告独立重复次数、每组干面质量、盲评流程或独立感官评审人数。评分为作者的结构化主观判断，未进行统计检验。实验也未测量 pH、仪器质构、含水率、煮制损失或仪器色差。照片可能受到光线、曝光、相机处理和拍摄时间影响。对照组虽在原帖中提及，但没有单独评分。因此，本结果应理解为可复现性有待进一步验证的实用厨房观察，而非普遍性结论。

This was a single-observer exploratory kitchen trial. Independent replication, dry pasta mass per treatment, blinding, and an external sensory panel were not documented. Scores were structured subjective judgments without statistical testing. pH, instrumental texture, moisture, cooking loss, and calibrated color were not measured. Photography may have been affected by lighting and camera processing. A control was mentioned but not independently scored. The findings are practical observations that require replication rather than universal conclusions.

7. 结论 / Conclusion

在本次 3 × 3 厨房实验中，4 g/L 小苏打、煮制 15 分钟获得最高综合评分 9.5/10；4 g/L、12 分钟则提供更偏硬的选择。2 g/L 改变有限，6 g/L 尤其在长时间煮制后产生苦涩、黏口和肥皂味。对希望利用商品意大利面制作拉面风格替代品的家庭烹饪者而言，4 g/L、15 分钟可作为透明且易操作的起始方法。

In this 3 × 3 kitchen trial, 4 g/L baking soda for 15 minutes received the highest score, 9.5/10, while 4 g/L for 12 minutes offered a firmer alternative. The 2 g/L treatments produced limited transformation, and 6 g/L, particularly with prolonged cooking, produced bitterness, stickiness, and a soap-like odor. For home cooks seeking a ramen-style substitute using commercial spaghetti, 4 g/L for 15 minutes is a practical starting method.

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数据可用性 / Data availability

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探索性烹饪比较报告 • Exploratory Culinary Comparison Report

五种商品意大利面在碱水煮制下的尺寸变化、时间进程与中式面条适配性：一项探索性厨房比较

Dimensional Changes, Time Course, and Chinese-Noodle Applications of Five Commercial Pasta Types Cooked in Alkaline Water: An Exploratory Kitchen Comparison

作者 / Author: 留子也要好好吃面 • Liuzi Ye Yao Haohao Chi Mian

文章编号 / Article ID: NN-2026-002　发表 / Published: October 2026



图 1 / Figure 1. 第二篇研究的成品呈现与 2.8 mm 熟面测量示例。 / Finished-dish presentation and an example cooked measurement of 2.8 mm.

摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房比较在统一条件（1 L 沸水加入 4 g 小苏打）下，测试 Angel Hair、Thin Spaghetti、Spaghetti、Linguine 和 Fettuccine 五种商品意大利面。作者记录达到“刚好弹牙”时的建议煮制时间，并用卡尺比较生面与熟面尺寸。五种面条的建议时间为 5、9、15、17 和 18 分钟；对应生面/熟面尺寸分别为 1.2/1.8、1.4/2.1、1.8/2.8、3.1/4.6 和 4.6/6.8 mm。平均尺寸膨胀倍数为 1.50。天使细面时间序列显示，3-8 分钟直径由 1.5 增至 1.9 mm，但 5 分钟被选为感官端点。结果提示面条规格、几何形状与感官熟度共同影响实用煮制时间。

关键词： 意大利面；小苏打；碱水煮制；卡尺测量；尺寸膨胀；中式面条替代

Abstract: Five commercial pasta types - Angel Hair, Thin Spaghetti, Spaghetti, Linguine, and Fettuccine - were compared under a standardized kitchen condition of 4 g baking soda in 1 L boiling water. Recommended cooking times at a subjectively al dente endpoint were 5, 9, 15, 17, and 18 minutes. Reported dry/cooked dimensions were 1.2/1.8, 1.4/2.1, 1.8/2.8, 3.1/4.6, and 4.6/6.8 mm, producing a mean dimensional expansion ratio of 1.50. An Angel Hair time series increased from 1.5 mm at 3 minutes to 1.9 mm at 8 minutes, although 5 minutes was selected as the preferred sensory endpoint. Pasta size, geometry, and sensory readiness jointly influenced practical cooking time.

Keywords: pasta; baking soda; alkaline cooking; caliper measurement; dimensional expansion; Chinese noodle substitute

1. 背景 / Background

第一篇研究确定 4 g/L 小苏打是标准意面的实用起点。第二篇进一步横向比较五种规格，以回答不同意大利面应煮多久、煮后尺寸如何变化，以及适合哪些中式面条应用。 / The first study identified 4 g/L as a practical condition for standard spaghetti. This second study compared five formats to examine cooking time, dimensional change, and possible Chinese-noodle applications.

2. 材料与方法 / Materials and Methods

五种商品意大利面均由作者自行购买。统一使用 1 L 沸水和 4 g 小苏打。达到作者判断的“刚好弹牙”状态后记录时间，并以数显卡尺测量。圆形面记录直径，Linguine 与 Fettuccine 记录宽度。 / Five commercial products were independently purchased. Each was cooked in 1 L boiling water with 4 g baking soda. Time was recorded at the author-defined al dente endpoint. A digital caliper was used; diameter was recorded for round strands, while width was recorded for Linguine and Fettuccine.



图 2 / Figure 2. 本研究所用五种商品意大利面。品牌仅用于材料识别，不构成赞助或认可。 / Five commercial pasta products used. Branding identifies materials only and does not indicate sponsorship or endorsement.

3. 五种面条的比较结果 / Comparative Results

类型 / Type	生面 / Dry (mm)	熟面 / Cooked (mm)	倍数 / Ratio	时间 / Time
Angel Hair	1.2	1.8	1.50×	5 min
Thin Spaghetti	1.4	2.1	1.50×	9 min
Spaghetti	1.8	2.8	1.56×	15 min
Linguine*	3.1	4.6	1.48×	17 min
Fettuccine*	4.6	6.8	1.48×	18 min

* Linguine 与 Fettuccine 的数值为宽度；三种圆形面为直径。不同几何量不能被解释为完全等价的截面变化。 / Linguine and Fettuccine values are widths; the round products are diameters. These geometries are not equivalent cross-sectional measures.

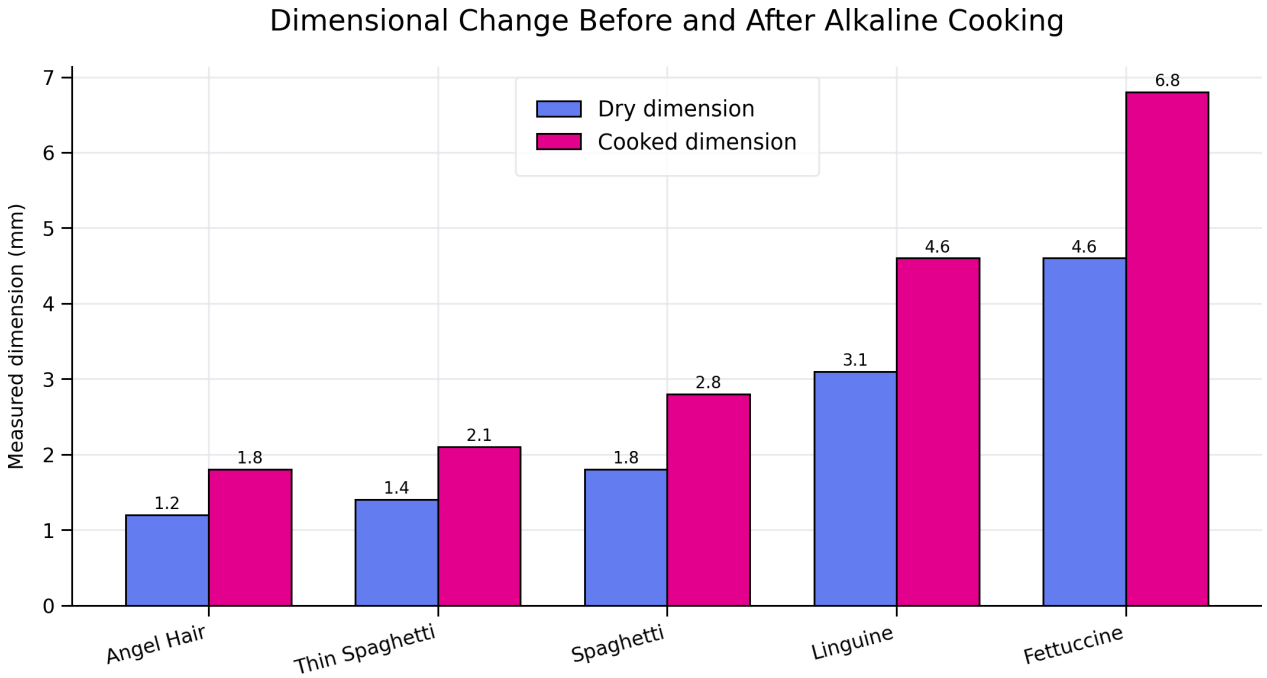


图 3 / Figure 3. 生面与熟面测量尺寸。 / Reported dimensions before and after alkaline cooking.

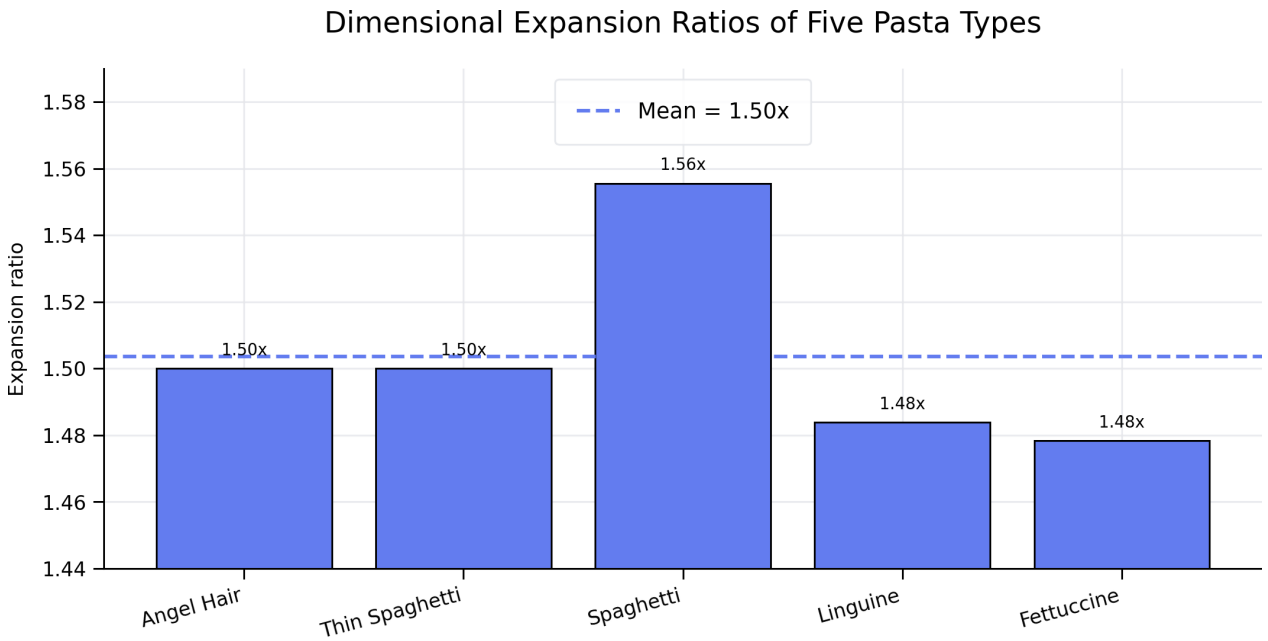


图 4 / Figure 4. 五种面条的尺寸膨胀倍数，平均 1.50 倍。 / Dimensional expansion ratios; mean 1.50×.

4. Angel Hair 时间进程 / Angel Hair Time Course



图 5 / Figure 5. 3-8 分钟的 Angel Hair 卡尺读数：1.5、1.6、1.7、1.8、1.8、1.9 mm。 / Caliper readings from 3 to 8 minutes.

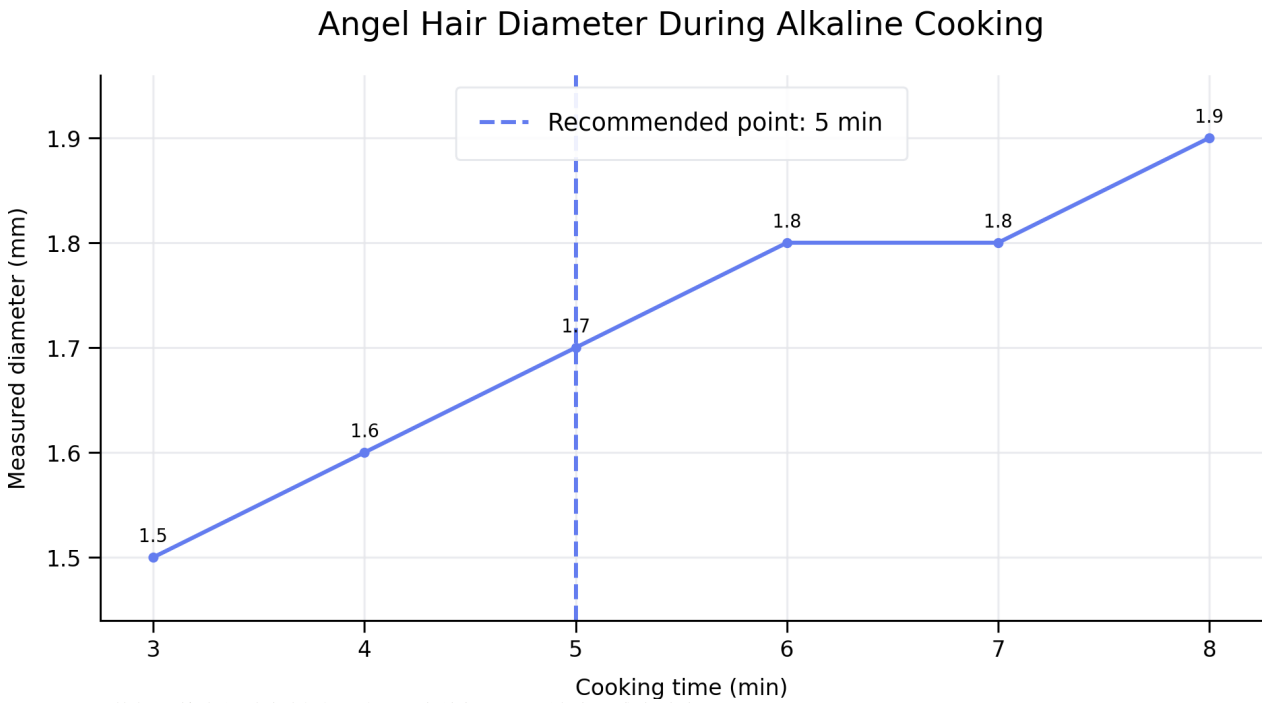


图 6 / Figure 6. 天使细面的直径随煮制时间增长；虚线标识 5 分钟建议感官端点。 / Angel Hair diameter increased with time; the dashed line marks the 5-minute recommended sensory endpoint.

3 分钟时图像仍显示硬芯。虽然 6-8 分钟继续膨胀，但原始观察指出超过约 6 分钟容易偏软，说明最大尺寸并不等于最佳口感。 / A firm core was still visible at 3 minutes. Continued swelling through 8 minutes did not imply optimal texture; the source notes softening beyond about 6 minutes.

5. 规格与建议时间 / Size and Recommended Time

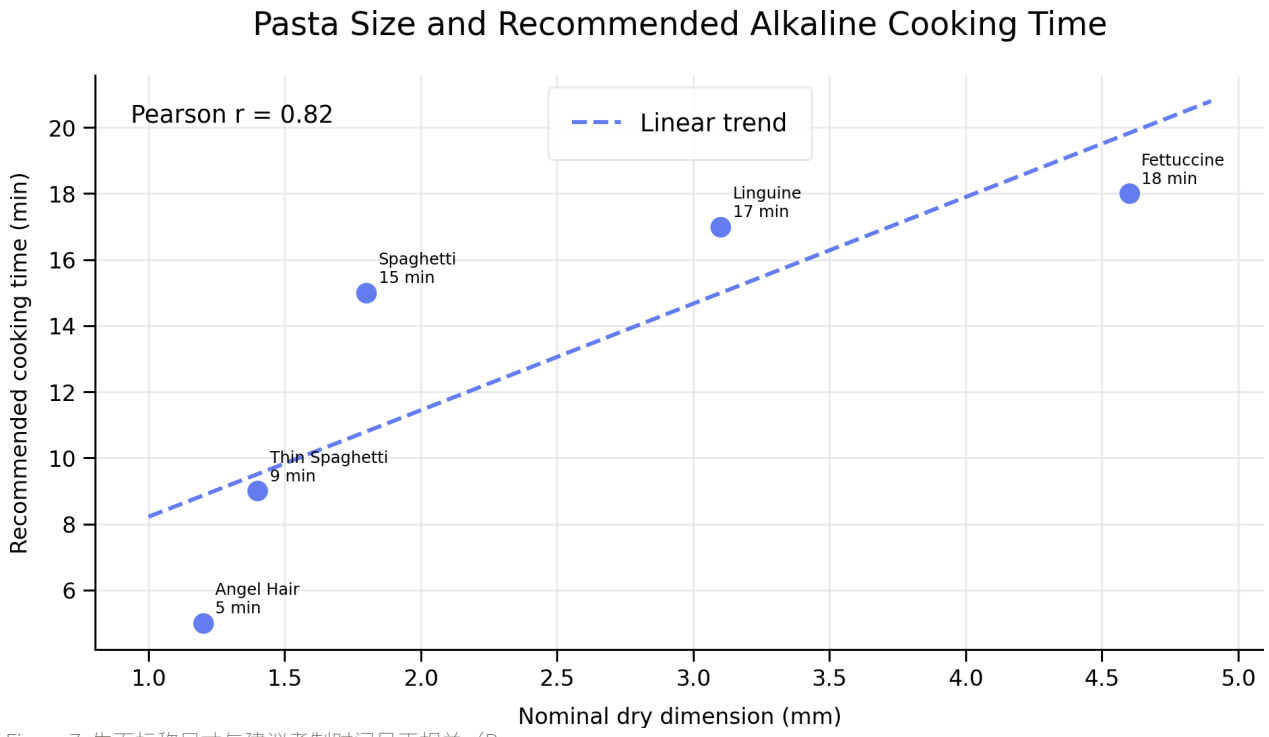


图 7 / Figure 7. 生面标称尺寸与建议煮制时间呈正相关（Pearson r = 0.82），但样本仅五种且几何测量并不完全等价，因此该相关仅为描述性结果。 / Nominal size was positively associated with recommended time (Pearson r = 0.82), but the five-item, mixed-geometry dataset supports descriptive interpretation only.

标准 Spaghetti 被原始材料标注为适用范围较广；Angel Hair 与 Thin Spaghetti 更适合细面应用；Linguine 与 Fettuccine 更适合炒面、拌面和宽面风格。建议属于作者的烹饪判断，不表示这些产品与对应中式面条在原料或加工结构上完全相同。 / Standard Spaghetti was judged the most versatile. Finer formats suited thin-noodle applications, while Linguine and Fettuccine suited stir-fried, mixed, or broad-noodle styles. These are culinary suggestions, not claims of structural identity.



图 8 / Figure 8. 原始五种面条结果与应用建议。 / Original five-pasta results and application suggestions.

6. 讨论 / Discussion

五种产品的膨胀倍数集中于
1.48-1.56，说明在本次条件下，尺寸相对增长较为一致；但绝对增长和煮制时间随规格明显增加。Angel Hair 时间序列进一步显示，尺寸增长可在感官最佳点后继续发生。因此，家庭操作应同时考虑卡尺读数、面芯状态和口感，而不能只追求更大尺寸。 / Expansion ratios clustered from 1.48 to 1.56, while absolute growth and cooking time increased with product size. The Angel Hair series shows that swelling continued beyond the sensory endpoint, so caliper values, center condition, and texture should be considered together.

7. 局限性 / Limitations

目前材料未说明每种面条的测量根数、独立重复次数、卡尺校准、测量位置与夹持力度。熟面柔软，卡尺可能造成压缩。Linguine 和 Fettuccine
为宽度，不能与圆面直径直接等同。建议时间基于单一作者的感官判断；Pearson r = 0.82
来自五个异质产品，仅用于描述，不用于统计推断。 / Strand count, replication, caliper calibration, measurement location, and jaw pressure were not documented. Soft cooked strands may compress. Width measurements for flat pasta are not directly equivalent to round-pasta diameter. Recommended times reflect one author sensory endpoint. The correlation of r = 0.82 is based on five heterogeneous products and is descriptive, not inferential.

8. 结论 / Conclusion

在 4 g/L 小苏打条件下，五种意大利面的建议煮制时间为 5-18 分钟，测量尺寸平均约增至生面的 1.50 倍。规格越大，通常需要更长时间，但几何形状与感官端点同样重要。该数据可作为家庭烹饪的起始参考，仍需重复测量和更标准化的质构评价。 / Under 4 g/L baking soda, recommended times ranged from 5 to 18 minutes and measured dimensions averaged approximately 1.50 times the dry values. Larger formats generally required longer cooking, while geometry and sensory endpoint also mattered. The dataset offers a practical starting reference that warrants replication and standardized texture testing.

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